CLEANING AND PREPARING THE DATA

I did not found missing data or duplicate. The file 0an1.csv in the same directory – this is the same data, but with transform into 0 and 1, for eventual machine learning.

Exploratory Data Analysis (EDA)

Churn rate:

Contract type:

Contract	total_customers	churned_customers	churn_rate_percent
Month-to-month	3875	1655	42.71
One year	1473	166	11.27
Two year	1695	48	2.83

It looks like that the churn rage by contract is as follows:

- Month-to-month contract – 3875 people total and 1655 of them are in churn, which is 42.71% of all of them;
- One year contract – 1473 people total and 166 of them are in churn, which is 11.27% of all of them;
- Two-year contract – 1695 people total and 48 of them are in churn, which is 2.83% of all of them.

It looks like that the people with shorter contract is more likely to withdraw!

1. Month-to-month contract:

Percentage Distribution of Services Among Churned Month-to-Month Customers

1655 people total in churn:

	Service	Yes_Percent	No_Percent	Total_Customers
▶	OnlineSecurity	12.9	81.1	1655
	OnlineBackup	24.5	69.5	1655
	DeviceProtection	25.2	68.8	1655
	TechSupport	12.4	81.6	1655
	StreamingTV	39.7	54.3	1655
	StreamingMovies	39.6	54.4	1655
	PaperlessBilling	75.5	24.5	1655

The people with such contract are in churn, because mainly of bad or no Online security (81.1% of them), Online backup (69.5% of them), Device Protection (68,8% of them) and Tech Support (81.6% of them). Still Streaming TV and Streaming Movies is also high percent – more than half of the people.

Recommendation: It is possible the reason to be not enough workers or perhaps technical problems or maybe the those people are not trained enough and they make some mistakes. You should check which is the exact reason. Only then you will know what exactly should be done for improving and reducing the number of the clients that you are losing.

2. One-year contract:

Percentage Distribution of Services Among Churned One-Year Contract Customers

166 people total in churn:

Service	Yes_Percent	No_Percent	Total_Customers
OnlineSecurity	36.1	58.4	166
OnlineBackup	55.4	39.2	166
DeviceProtection	56.6	38.0	166
TechSupport	45.2	49.4	166
StreamingTV	72.9	21.7	166
StreamingMovies	77.1	17.5	166
PaperlessBilling	71.1	28.9	166

The people with one-year contract are in churn, because of problems with the Online Security (58.4%), also Tech Support (49.4). Some people are not happy enough, because also of the Online Backup (39.2%) and Device Protection (38.0%).

Recommendation: The same thing: Maybe because of not enough workers, technical problems or not well enough trained workers.

3. Two-year contract:

Percentage Distribution of Services Among Churned One-Year Contract Customers

48 people in churn:

Service	Yes_Percent	No_Percent	Total_Customers
OnlineSecurity	45.8	43.8	48
OnlineBackup	54.2	35.4	48
DeviceProtection	70.8	18.8	48
TechSupport	60.4	29.2	48
StreamingTV	75.0	14.6	48
StreamingMovies	70.8	18.8	48
PaperlessBilling	68.8	31.3	48

People with Two-Year contract in churn. They are maybe more pretentious. The NO percents are not that high. Online security with 43.8%, Online Backup – 35.4%, Paperless Billing – 31.3%.

Payment method:

PaymentMethod	total_customers	churned_customers	churn_rate_percent
Electronic check	2365	1071	45.29
Mailed check	1612	308	19.11
Bank transfer (automatic)	1544	258	16.71
Credit card (automatic)	1522	232	15.24

Here is the churn rate of the people by Payment method. According to this data part, it says that mostly the people using the Electronic check method – 1071 people of 2365 or 45.29% of all of them. Followed by the Mailed check – 19.11%. Then is Automatic Bank Transfer – 16.71% and finally – Automatic Credit Card – 15.24%.

1. **Electronic Check – 1071 people in churn**

Service	Yes_Percent	No_Percent	Total_Customers
OnlineSecurity	12.5	86.1	1071
OnlineBackup	26.7	71.9	1071
DeviceProtection	28.9	69.7	1071
TechSupport	12.7	85.9	1071
StreamingTV	48.4	50.2	1071
StreamingMovies	47.6	51.0	1071
PaperlessBilling	81.0	19.0	1071

The people using the Electronic Check payment method complain mostly about the Online Security – 86.1% of all 1071, which is a high percent.

In second place we have with 85.9% - Tech Support – high percent.

In third place is the Online Backup – 71.9% - high percent.

2. Mailed Check – 308 people in churn

	Service	Yes_Percent	No_Percent	Total_Customers
▶	OnlineSecurity	13.3	63.6	308
	OnlineBackup	16.9	60.1	308
	DeviceProtection	18.5	58.4	308
	TechSupport	16.6	60.4	308
	StreamingTV	22.4	54.5	308
	StreamingMovies	20.1	56.8	308
	PaperlessBilling	58.1	41.9	308

The people using the Mailed check payment method complain mostly about the Online Security – 63.6%, followed by Tech Support – 60.4% and Online Backup – 60.1%

3. Bank Transfer (automatic) – 258 people in churn

	Service	Yes_Percent	No_Percent	Total_Customers
▶	OnlineSecurity	22.1	70.9	258
	OnlineBackup	34.9	58.1	258
	DeviceProtection	37.2	55.8	258
	TechSupport	21.7	71.3	258
	StreamingTV	48.1	45.0	258
	StreamingMovies	50.4	42.6	258
	PaperlessBilling	72.1	27.9	258

About the people in churn using the Bank Automatic Transfer – we have most complains about the Tech Support – 71.3%, Online Security – 70.9%, also Online Backup – 58.1% and Device Protection with 55.8%.

4. Credit Card (automatic) – 232 people in churn

Service	Yes_Percent	No_Percent	Total_Customers
OnlineSecurity	27.2	69.0	232
OnlineBackup	40.9	55.2	232
DeviceProtection	35.8	60.3	232
TechSupport	28.9	67.2	232
StreamingTV	44.4	51.7	232
StreamingMovies	50.0	46.1	232
PaperlessBilling	72.4	27.6	232

We have most complains about the people using Credit Card (automatic) method in the Online Security – 69.0%, followed by Tech Support – 67.2% and Device Protection 60.3%.

Tenure:

tenure_group	total_customers	churned_customers	churn_rate_percent
0-12 months	2186	1037	47.44
13-24 months	1024	294	28.71
25-48 months	1594	325	20.39
49+ months	2239	213	9.51

Here is the churn rate of the people by Tenure. Looks like that the people with the shortest tenure are with highest number of churned customers. (47.44%)

Those with the biggest tenure are hardly withdrawing. (213 people, only 9.51%)

1. Tenure group (0-12 months – 1037 people)

Service	Yes_Percent	No_Percent	Total_Customers
OnlineSecurity	8.9	82.3	1037
OnlineBackup	15.7	75.4	1037
DeviceProtection	17.7	73.4	1037
TechSupport	10.3	80.8	1037
StreamingTV	30.2	60.9	1037
StreamingMovies	29.9	61.2	1037
PaperlessBilling	71.3	28.7	1037

Here we have the first tenure group (0-12 months) – according to this part of the data, we have 82.3% of them that quit, because of the Online Security. We also have high percent of withdraw thanks to the Tech Support – 80.8% Other high percents are related to the Online Backup (75.4%) and Device Protection (73.4%).

2. Tenure group (13-24 months – 294 people)

Service	Yes_Percent	No_Percent	Total_Customers
OnlineSecurity	19.7	77.2	294
OnlineBackup	27.9	69.0	294
DeviceProtection	32.0	65.0	294
TechSupport	17.3	79.6	294
StreamingTV	49.7	47.3	294
StreamingMovies	49.7	47.3	294
PaperlessBilling	80.6	19.4	294

Second tenure group (13-24 months) – most complains and therefore withdraws, because of the Tech Support (79.6%), Online Security (77.2%), Online Backup (69.0%) and Device Protection (65.0%).

3. Tenure group (25-48 months – 325 people)

Service	Yes_Percent	No_Percent	Total_Customers
OnlineSecurity	23.1	75.1	325
OnlineBackup	43.7	54.5	325
DeviceProtection	43.7	54.5	325
TechSupport	25.5	72.6	325
StreamingTV	57.5	40.6	325
StreamingMovies	60.0	38.2	325
PaperlessBilling	80.3	19.7	325

Third tenure group (25-45 months) – most complains because of Online Security (75.1%) and Tech Support (72.6%). Also more than half of the people are negative, because of the Online Backup and Device Protection (54.5% each).

4. Tenure group (49+ months – 213 people)

Service	Yes_Percent	No_Percent	Total_Customers
OnlineSecurity	32.9	64.3	213
OnlineBackup	63.8	33.3	213
DeviceProtection	58.7	38.5	213
TechSupport	32.4	64.8	213
StreamingTV	78.9	18.3	213
StreamingMovies	78.4	18.8	213
PaperlessBilling	76.5	23.5	213

Forth tenure group (49+ months) – most complains because of the Tech Support (64.8%) and Online Security (64.3%).

There are not many people that are withdrawing maybe because the longerstayed community is not so pretentious. They

TotalCharges_Range	Churn_Percent	Total_Customers
< 500	41.2	2011
> 2000	18.6	2856
1000–2000	21.0	1283
500–1000	27.0	893

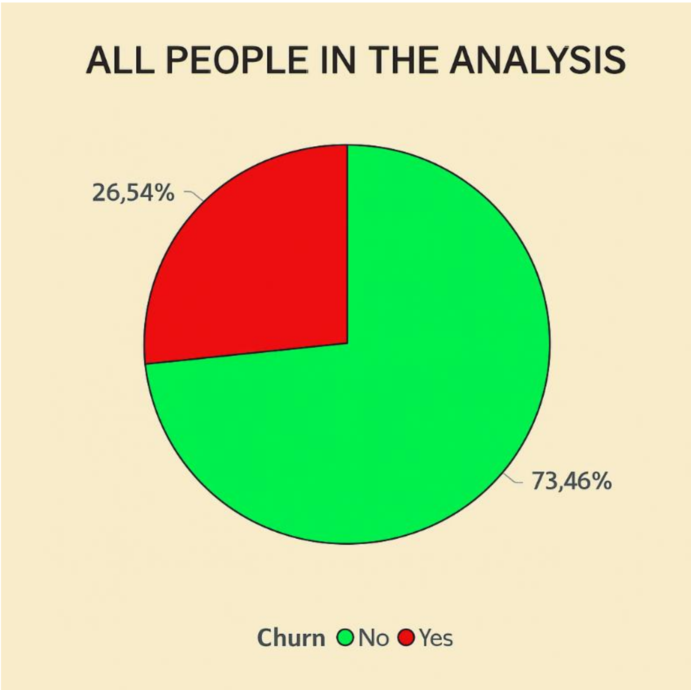
Churn	Avg_TotalCharges	Customer_Count
No	2549.91	5174
Yes	1531.8	1869

These tables show that the the people who pay more is less likely to pull away. It seems that those who have paid less than 500 are higher drop out percent. Those who pay more than 2000 are under 20%.

The average payment (second table) is higher for those who remain and they are almost 3 times more in quantity than those who pull away.

Correlations between features and churn

Variable	Correlation with Churn	Interpretation
`tenure`	**-0.354**	Customers who have been with the company longer are significantly less likely to churn. The negative correlation indicates that as tenure increases, the likelihood of churn decreases.
`MonthlyCharges`	**+0.193**	Customers with higher monthly charges show a slight tendency to churn more. The positive correlation suggests that higher costs may contribute to dissatisfaction or cancellation.
`TotalCharges`	**-0.199**	Customers who have paid more in total are less likely to churn. This may reflect longer relationships or higher overall engagement with the service.



I made an analysis of all the people in the data. As we can see, most of the people do not have many complaints. They prefer to remain, having such service. Around a quarter of them have decided to withdraw.

Customer Value = Monthly Charges × Tenure

I have it in the file called CustomerValue, which is in the same directory or click on this icon here (link), or in the Github Portfolio. (because it is too big)

CustomerValue.csv

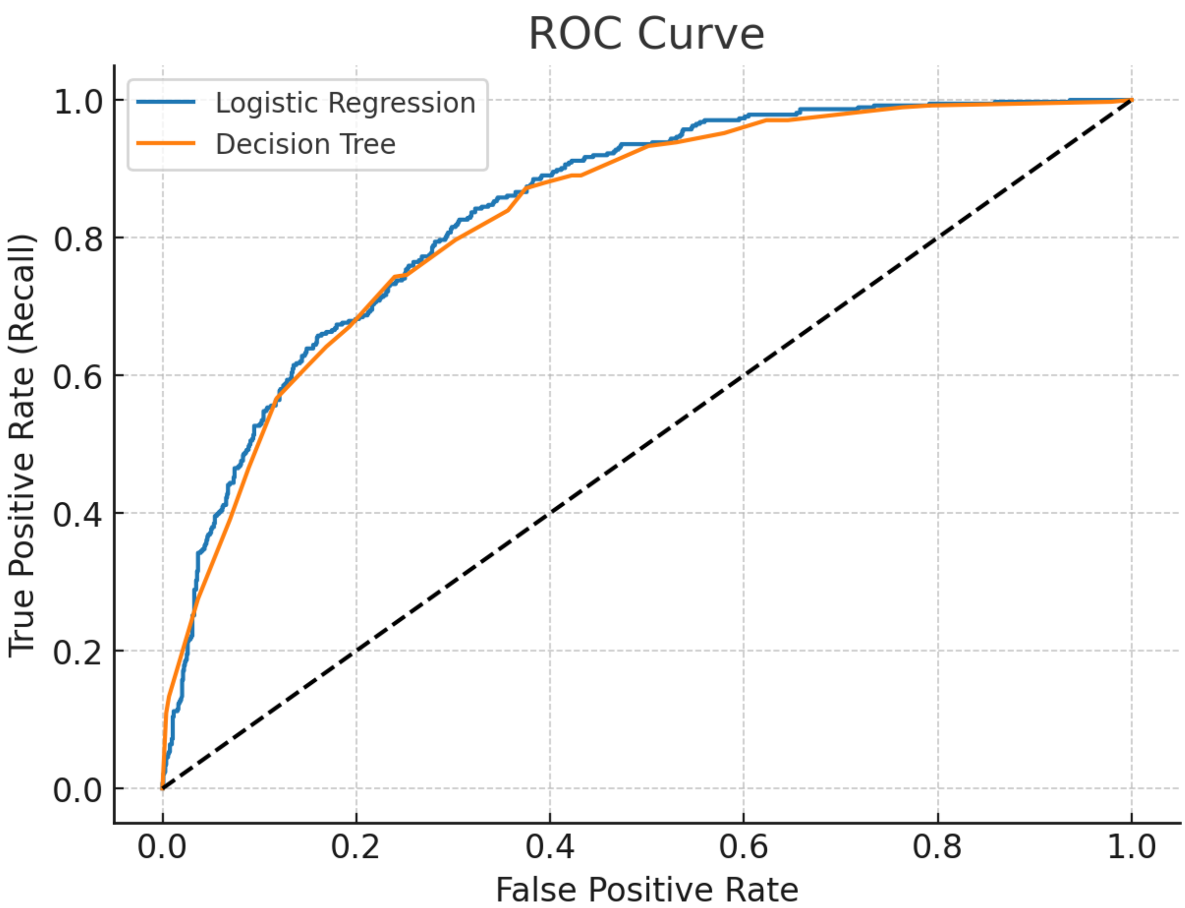
Grouping customers by risk level based on behavior

RiskLevel.csv

Same thing – In the Directory, Github or click on the link.

Predictive Modeling

Building a logistic regression / decision tree model to predict churn. Evaluating model performance using accuracy, precision, recall, and ROC curve



Model	Accuracy	Precision	Recall	F1-score	ROC AUC
Logistic Regression	0.803407	0.652038	0.556150	0.600289	0.842639
Decision Tree	0.798439	0.634731	0.566845	0.598870	0.830278

Logistic Regression Model (*The model's ability to distinguish between churned and non-churned customers. The closer the curve is to the top-left corner, the better the model's performance.*):

- Accuracy – a bit more than 80% of all people are correctly predicted;
- Precision – not much hopeful predictions by the model – 65%;

- Recall – not much of capturing on the withdrawn people – 55%;
- F1-score (balance between Precision and Recall) – 60% - not very good, close to average. Mistakes done by the model;
- ROC AUC – about making a difference between the people that have pulled off and those who did not – it's 84% - so good, but still need more work upon it, if it is going to be improved.

Decision Tree Model (*The branches of the model that lead to churn predictions. Each node represents a condition, and the leaves show the final decision.*):

- Accuracy – almost 80% of the people predicted;
- Precision – 63%;
- Recall – 57% of the people are caught as withdrawn;
- F1-score – 60%;
- ROC AUC – 83% the tree model makes a difference between pulled away and not pulled away clients.

Business conclusions and recommendations

The best thing that can be done is improving the resources (materials and workers). To see what causes people to quit from the services and to find the correct and possibly the best way to make it better. It is better for the business to be more budgeted and that way, which gives a better opportunity for growing up and success. You can also make a review space where you can see the opinion of the different people and making a segment opportunity for them to choose from which category they are (longer or shorter tenure group, which payment method are they using or what contract). That way you can easily find out what reforms you need to do to avoid falling down.

Another tactic that may help is to increase the salary of the more effective workers and give them the word if they complain about something related to their job or possible discounts if you see disadvantages, especially known by the clients.

There are 3 possible important moments, which make client losing:

- Not enough workers;
- Problems with business resources (not enough people or ineffective materials;
- Lack of review system – to find out easily what could be the negative moments of the activity.

Situation	**Result**
Too much work and not enough workers.	*Workers getting tired and quality it falling down.*
Ineffective material resources.	*Additional worker commitment.*
No good review system.	*Cannot find out well enough the opinion of the clients and what do they see.*

This analysis is created by:

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During the analysis the following platforms and programs were used:

- *Microsoft Word;*
- *Microsoft Excel;*
- *Microsoft Power BI;*
- *My SQL;*
- *AI (sometimes when I have difficulties for making a chart or go to an exact option)*